Supervised Learning & Model Engineering
Diabetic Patients' Readmission Prediction



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Philomène CHOLET
Kelsy NGUYEN
Inès SIMOND-CHARBINAT

EDA & PROBLEM FRAMING

DATASET – Understanding and problem formulation

We frame the task as an encounter-level binary classification: at discharge, predict whether a diabetic patient will be readmitted within 30 days.
The intended use is follow-up prioritisation: ranking stays so that a limited follow-up capacity reaches the highest-risk patients first.
This study evaluates prediction only; it does not show that follow-up prevents readmissions.

Hospital Stays (rows)	101,766
Patients	71,518
Source	130 hospitals / 1999-2008
Columns	49 (2 identifiers + 47 candidates)
Target	Readmission (column 50)
Duplicated Rows	0

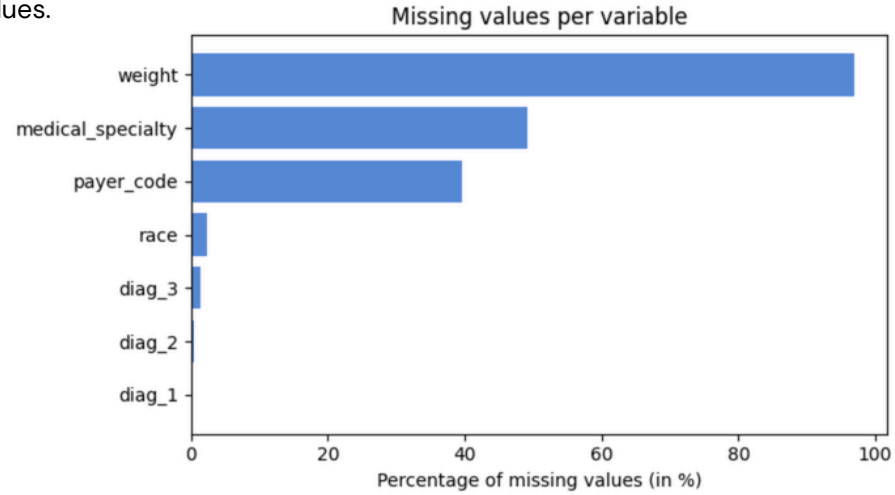
Hence the question is: ***At discharge, using only the information available at that moment, which diabetic patients should be followed up first so that a limited follow-up capacity can reach as many 30-day readmissions as possible?***

Original Class (readmission)	Share of stays	readmitted_30d	Meaning
NO	53.91%	0	No readmission recorded
>30	34.93%	0	Readmitted after 30 days
<30	11.16%	1	Readmitted within 30 days (positive class)

Our prediction-time rule requires every input to be available at discharge. The future outcome is excluded from predictors, as are encounter_id and patient_nbr.
The latter remains available for grouping. Race is excluded from model inputs and retained separately for the fairness audit; its removal alone can't guarantee fairness.

EDA – Cohort definition and sampling

The identified substantial missingness: weight (96.9%) was removed, medical_specialty (49.1%) and payer_code (39.6%) were retained with an explicit missing category, and question marks were normalised to missing values.



Partition	Encounters	Readmission rate
Training (develop-ment)	79,625	11.46%
Testing	19,718	11.10%

Administrative *_id fields were handled as categories. 15 medication columns were quasi-constant, but all medication inputs were retained; one-hot encoding groups categories observed fewer than 50 times in the fitting data.
Encounters with discharge codes 11, 13, 14, 19, 20, or 21 were excluded as death/hospice stays: 2,423 encounters (2.4%). This defines the scope of the proposed follow-up workflow. Hospice exclusion is not evidence that readmission or follow-up is impossible. The retained cohort contains 99,343 encounters with 11.39% positive outcomes.
Because 23.5% of patients had repeated encounters, StratifiedGroupKFold was used to reserve approximately 20% for testing, with zero patient overlap.

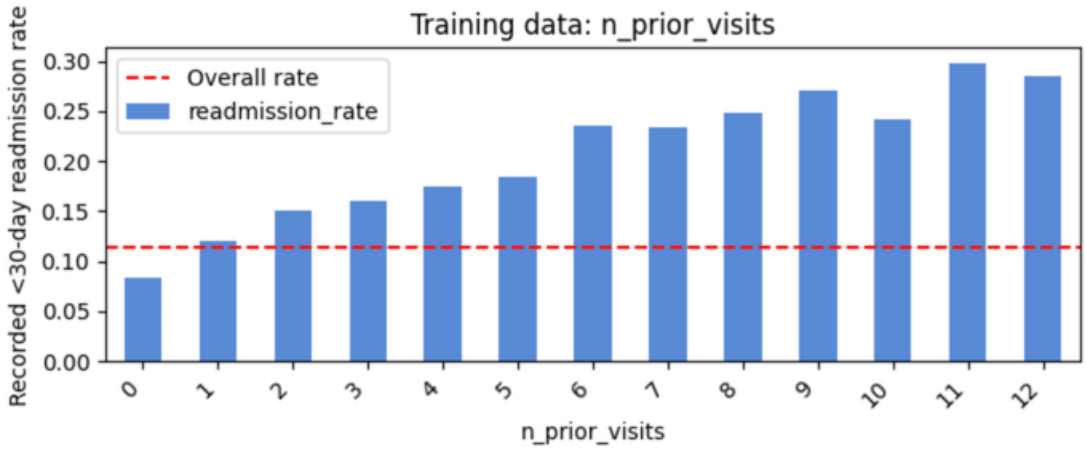
FEATURE ENGINEERING

FEATURES – Engineering and preprocessing

Engineering reorganises existing discharge-time information into interpretable representations.
The row-wise add_features function is the first pipeline step. Its safety depends on source availability at prediction time, not merely on being row-wise.

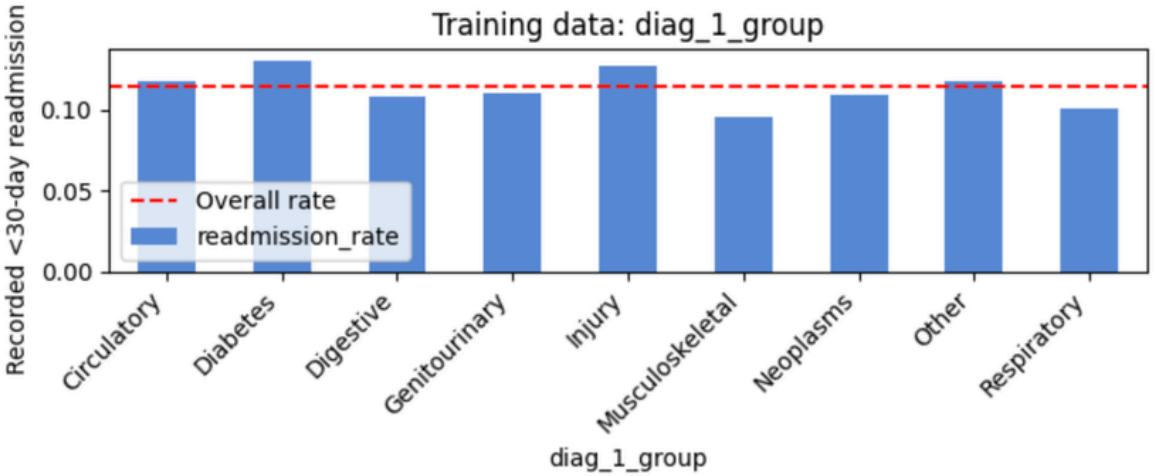
Feature	Construction	Why?
n_prior_visits	Outpatient + emergency + inpatients visits in the past year	One summary of prior care stays
n_med_changes	Number of the 23 diabetes medications whose dose went up or down	Treatment instability
age_years	Middle of the 10-year bracket (e.g., [70-80] → 75)	Keeps the order of the brackets as a number
diag_1/2/3_group	700+ ICD-9 codes → turned into 9 groups (Strack et al., 2014) V/E codes → Other; missing → Missing	Sparse codes become readable groups

The diagnosis mapping follows the grouping attributed to Strack et al. in the notebook. Special V/E codes are mapped to Other, missing codes to Missing, and malformed numeric inputs raise an error. The final representation has 11 numeric and 36 categorical inputs before encoding.
Median imputation and standard scaling are applied to numeric inputs; missing-category imputation and one-hot encoding are applied to categoricals.
A ColumnTransformer and Pipeline ensure learned preprocessing is refitted inside each training fold.



Number of med changes on training data

Number of med changes	Number of stays	Readmission rate
0	57,997	0.107
1	20,504	0.135
2	1,033	0.127



CLASS IMBALANCE HANDLING

METRICS – How will we evaluate the models

Before handling the train-test split and the different models, it is important to first determine the different metrics we will look at to evaluate the different models and methods used over a 5-fold cross-validation on the training set, grouped by patient and stratified (all models use the same folds).

- Recall is the share of readmitted patients that is flagged (1). Missing them is the costly error for hospitals;
- Precision is the share of flagged patients really readmitted;
- F1 is the balance between the precision and recall scores;
- ROC-AUC is the quality of the ranking (0.5 is taken at random to start with);
- PR-AUC is the ranking quality focused on the rare class (random around 0.11). This is the main metric.

WEIGHTS – Are class weights necessary?

In order to compute the weight of class balancing, three tests are performed on the training set (*cf. appendix class_weights_test_results*):
0.5 is taken as an arbitrary threshold.

- The dummy model: this model gets 88.5% accuracy, but the recall score is equal to 0, therefore, the accuracy is useless in this case. (The ROC-AUC is 0.492, not 0.5, because its constant score changes a little between folds.)
- A Logistic Regression without weights: this model ranks well, PR-AUC 0.216, twice the random level, but at threshold 0.5 it flags almost no patients, recall of 1.7%.
- Class weight balanced: the readmitted patients count about 7.7 times more, as the recall score goes up to 55.2% (precision 18.4%), but PR-AUC and ROC-AUC do not change. In this case, the balanced weights shift the scores, but they do not improve the ranking (*cf. appendix class_weights_graph*).

Without balanced weights, the best threshold is 0.122, and with weights, 0.524. Both give similar PR-AUC (0.216 vs 0.215) and ROC-AUC (0.663 vs 0.664). So class weights and threshold tuning do a similar task.

REMEDY – Is there a "miracle" remedy?

Having tested class weights, the question remains the same; therefore, four remedies are compared using a Logistic Regression model, on the same five folds, to determine which one would prove best:
- no remedy (baseline);
- class weights (previous test);
- random undersampling (remove majority readmissions);
- SMOTE (create synthetic readmitted patients' stays).

As for the results, three words: no real difference (*cf. remedy_tests_results*).
The PR-AUC scores of all four tests are 0.216 for the baseline, 0.216 for the class weights, 0.213 for undersampling, and 0.208 for SMOTE. It is possible to compute that all the differences are smaller than the fold standard deviation, around 0.010.

cf. remedy_tests_results

	PR-AUC	PR_AUC fold std	ROC-AUC	recall at 0.5	precision at 0.5	best F1 threshold	F1 at best threshold
baseline	0.216	0.009	0.663	0.017	0.481	0122	0.278
class weights	0.216	0.010	0.664	0.552	0.184	0.524	0.278
random undersampling	0.213	0.009	0.661	0.560	0.180	0.532	0.277
SMOTE	0.208	0.009	0.648	0.555	0.173	0.560	0.269

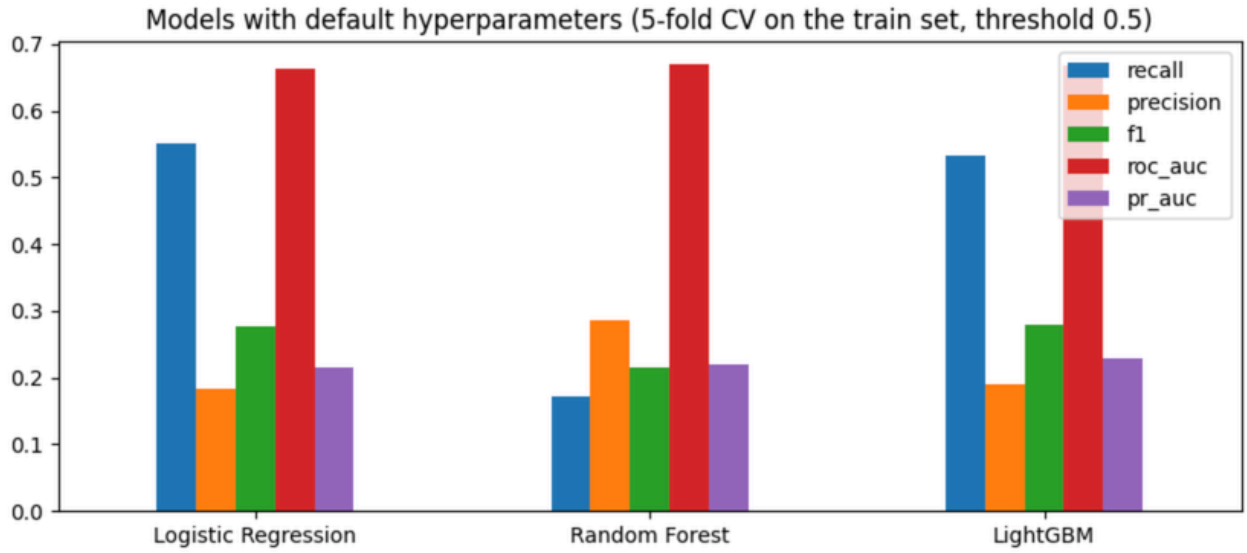
It was decided that balanced class weights will be used in the forward models' progression, as well as a specific threshold, 0.52, from the out-of-fold predictions.
The choice was made because class weights were the simplest, with no extra configuration step. The rule set was that another remedy is kept only if it beats class weights by more than the standard deviation between folds (0.0096).
In the end, balanced class weights do not improve the ranking, but they shift the scores in a way that the new chosen threshold is usable.

MODEL PROGRESSION & PIPELINE

MODELS – A three models' set-up

In order to predict the readmission of diabetic patients within 30 days, it is important to choose the correct model. For that, three models that work for categorical output will be put to the test on the training set:

- Logistic Regression: a simple linear baseline with 2 000 max iterations and a balanced class weight;
- Random Forest: the average of many decorrelated trees. The number of estimators is set at 300, with a max depth of 30, a minimum of 5 samples per leaf, and a balanced class weight;
- LightGBM: a gradient boosting model, with trees added one after another to correct the errors. The number of estimators is set at 300, with a learning rate of 0.5, 31 leaves, and a balanced class weight.



cf. models_training_results

	accuracy	balanced accuracy	precision	recall	F1	ROC-AUC	PR_AUC	PR_AUC std	train PR-AUC	fit time (s)
Logistic Regression	0.668	0.618	0.184	0.552	0.276	0.664	0.216	0.010	0.221	14.735
Random Forest	0.856	0.559	0.286	0.173	0.215	0.670	0.220	0.013	0.884	42.949
LightGBM	0.685	0.619	0.189	0.533	0.280	0.668	0.230	0.015	0.378	6.927

With the results above, it is possible to observe that:

- Logistic Regression: the PR-AUC is 0.216, and there seems to be almost no overfitting (train 0.221).
- Random Forest: the PR-AUC is 0.220, but there is strong overfitting (train 0.884), and it is the slowest.
- LightGBM: it has the best PR-AUC score, 0.230, is moderately overfitting (train 0.378), and is the fastest.

THE POTENTIAL CHOICE – Which model will probably be used forward

Something is sure: the Random Forest model will potentially be dropped, and the LightGBM model seems to be running for first place.

But these models have been trained on computable variables from the dataset, it is missing some variables made at the feature engineering part (e.g., age, number of previous visits, diagnosis groups, number of medication changes…).
An ablation is done with these new variables, where the start is from the raw columns and adding the new features one at a time, with the same folds. The PR-AUC scores are evaluated fold by fold (*cf. ablation_results*).

After this ablation, it is possible to see that all gains are within "noise" (at most +0.002 for Logistic Regression and +0.0032 for LightGBM, with a std of 0.002–0.004). It was decided that these additional features will be kept as they are "cheap", leakage-free, and easy to explain.

HYPERPARAMETER TUNING

OPTUNA – The search for the best parameters per model

As for the hyperparameter tuning part, Optuna is used as a helper to optimize the mean PR-AUC.
As Optuna runs trials, each one will be done using five-fold cross-validation of the whole pipeline, on the exact same folds, using the random seed 42. The first trial of each model will be the starting baseline configuration.

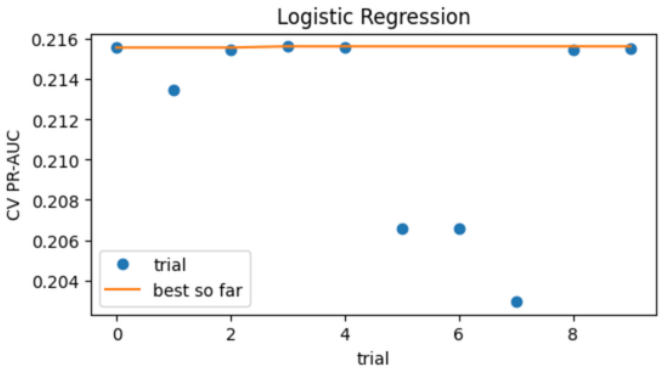
But what are the hyperparameters searched for each model?

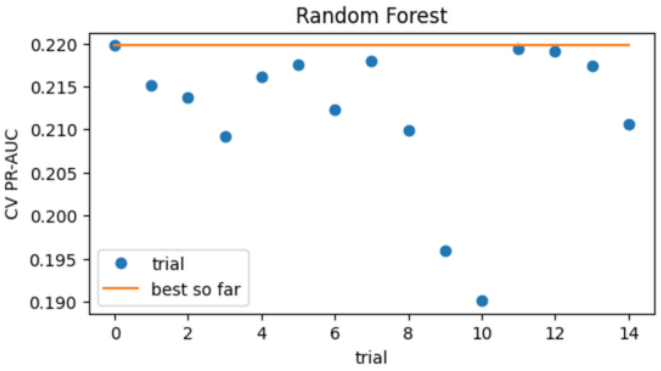
- Logistic Regression: "C";
- Random Forest: number of trees, depth, leaf size, and features per split;
- LightGBM: number of trees, learning rate, number of leaves, leaf size, sampling, and L2.

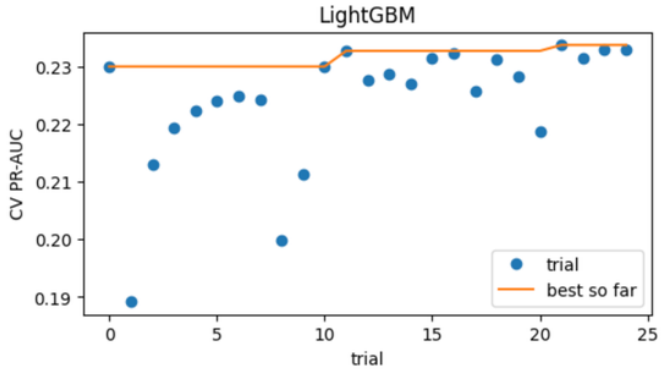
It is important to note that the best score that will be output will be a bit optimistic, as it was chosen on similar folds; of course, the test set stays untouched.

	default PR-AUC	best after tuning	gain
Logistic Regression	0.2156	0.2156	0.0001
Random Forest	0.2198	0.2198	0.0000
LightGBM	0.2301	0.2339	0.0037

cf. optuna_pr_auc_gain_comparison







The results over 56 runs (3 configurations and 50 trials):

- Logistic Regression (10 trials): its best cross-validation PR-AUC score is 0.2156; it took 10 minutes to compute, and the best hyperparameter is C = 0.46.
- Random Forest (15 trials): its best cross-validation PR-AUC score is 0.2198; it took 43 minutes, and the best hyperparameters are: 300 estimators, a max depth of 30, a minimum of 5 samples per leaf, and max features = "sqrt," which is the starting configuration.
- LightGBM (25 trials): its best cross-validation PR-AUC score is 0.2338; it took around 21 minutes, and the best hyperparameters are: 500 estimators (more trees), a smaller learning rate of 0.017, 56 leaves, and 58 stays per leaf. For this model, the gain stays smaller than the fold std (0.0037 vs 0.015).

With this hyperparameter tuning, it is possible to say that LightGBM remains the best model in the diabetic patients' early readmission case. However, it is important to keep in mind that the best score among many trials is slightly optimistic because it was selected on these same five folds. Nested cross-validation would correct this but costs about 5 times more to do, which is not ideal for healthcare institutions.

EXPERIMENT TRACKING

SUMMARY – The models

Comparing the three tuned models one last time with the same metrics, each at its best F1 threshold, it is possible to see that (*cf. tuned_models_comparison_results*):

	PR-AUC	F1	recall	precision	threshold
Logistic Regression	0.2156	0.278	48.9%	19.5%	0.524
Random Forest	0.2198	0.283	52.6%	19.4%	0.371
LightGBM	0.2301	0.291	44.6%	21.6%	0.542

LightGBM is the clear "winner" as it has the best PR-AUC and F1, but the differences are small.

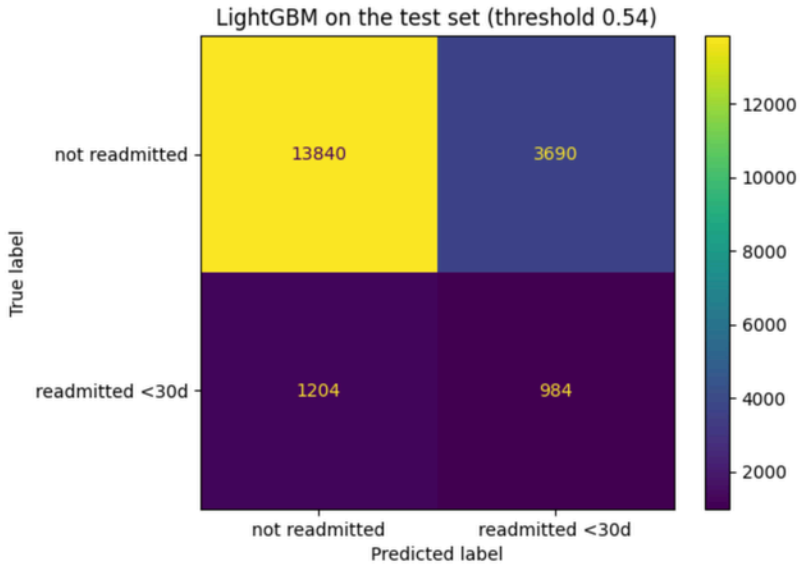
Now, overfitting needs to be checked on the training set only through a learning curve (*cf. learning_curve_lightgbm*).
With a set of 6 400 patients' visits, the training PR-AUC score decreases to 0.374, and the validation PR-AUC score increases to 0.234 (a standard deviation of 0.014). This means that there is moderate overfitting, much less than the Random Forest (0.884 vs 0.220). Tuning did not significantly reduce the gap (LightGBM: 0.378 vs 0.230). But it is possible to observe that as the number of patient visits increases, the two curves get closer, meaning that more data would help the gap decrease a bit.

TESTING – Bringing out the testing set

	accuracy	precision	recall	F1	ROC_AUC	PR_AUC
train (cross-validation)	0.751	0.216	0.446	0.291	0.676	0.233
test	0.752	0.211	0.450	0.287	0.682	0.235

The testing is done, but what do the results mean (*cf. test_set_results*)?

- The cross-validation PR_AUC score of the best trial is 0.234 for the train set and 0.235 for the test set (95% Confidence Interval: 0.213 to 0.258); the cross-validation estimate is inside the interval of the test score.
- The PR_AUC score on fresh training folds is 0.233 (std 0.012) compared to 0.234 on the folds used for tuning; therefore, the selection effect on the folds is +0.001, which is considered noise.



In conclusion, the final model chosen is the tuned LightGBM, with a threshold of 0.542. But in practice, the model finds 45% of the early readmissions, and 1 flagged patient in 5 is really readmitted (twice the base rate of 11.1%). Therefore, it can help prioritise follow-ups, but not act as a decision factor alone.

EXPLAINABILITY

PREDICTED RISK – What drives it?

We explain the tuned LightGBM with two methods. SHAP (TreeExplainer, 2,000 test samples) splits each prediction into feature contributions. We sum them per original variable, since one-hot encoding splits each categorical variable into many columns. Permutation importance (8,000 test stays, 5 repeats) measures the PR-AUC drop when a raw column is shuffled. SHAP shows what the model uses; permutation shows what its performance depends on.

Rank	SHAP: grouped per variable (mean / SHAP / log-odds)	Permutation (PR_AUC drop)
1	number_inpatients (0.235)	number_inpatients (around 0.077)
2	discharge_disposition_id (0.215)	discharge_disposition_id (around 0.050)
3	n_prior_visits (0.089)	diag_1 (around 0.008)
4	diag_1_group (around 0.082)	medical_specialty (around 0.006)
5	number_diagnoses (around 0.069)	payer_code (around 0.006)
6	medical_specialty (around 0.057)	number_emergency (around 0.005)

Both methods rank past inpatient stays and discharge destination first and second, far ahead of everything else, so this result is robust. Below them, permutation drops are under 0.01, and their order is not reliable. n_prior_visits ranks 3rd for SHAP but cannot appear in the permutation ranking because it is built inside the Pipeline. The ablation pointed the same way: the feature adds no new information but gathers the prior-care signal into one readable variable.

Long stays keep a high raw readmission rate in the EDA (12.7% at 14 days), yet SHAP gives them a negative effect. SHAP measures each effect given the other features: long stays already carry many diagnoses and medications. No top variable is known until after discharge, which is a further check against leakage.

Case	Score	Actual	Profile	Main Driver
Flagged	0.92	Readmitted	11 past stays, 3 emergency visits	number_inpatients +2.12
Missied	0.14	Readmitted	No past visit, 12-day stay, home discharge	All pushed down (long-stay -0.32)
Correct Low	0.06	Not readmitted	No past visit, 3-day stay, orthopaedic	medical_specialty -0.72

The missed case shows the model's limit. A patient with no history of care use looks low-risk, so readmissions driven by things that are not recorded, such as the home or social situation, stay invisible. This matches the modest PR-AUC (0.235, about twice the 11% base rate).

Business takeaways

- The model's logic is credible. It relies on known risk factors: prior use of care, discharge destination, and patient complexity. Clinicians can trust a flag and also challenge it.
- Act at discharge on two signals. Repeat inpatients and patients transferred to further care are the clearest follow-up priorities.
- Watch for blind spots. First-time patients are under-flagged, so the score should support clinical judgement, not replace it.
- The model shows association, not cause. It tells the team whom to follow up first, not which intervention prevents a readmission.
- Monitor data quality. A missing payer code raising the risk points to a recording effect, which should be checked before deployment.

CALIBRATION DIAGNOSTICS

CALIBRATION – The model is over-confident

A model is calibrated if its scores match real observed frequencies. Here, among patients who received a score of 0.3, about 30% should really be readmitted. Since a hospital could use this score as a risk level (for example, with an obligation to call every patient above a certain threshold), we checked the calibration of the retained model (LightGBM) on the test set.

To do so, we used 3 tools :

- Reliability curve: the test stays are split into 10 groups of equal size (about 1970 stays each) by predicted score. For each group, we compared the mean score with the observed readmission rate (Figure 1 : Reliability curve and distribution of predicted scores before calibration).
- Brier score: the mean squared error between the score and the real outcome (0 being perfect). As a reference, always predicting the base rate gives 0.0987.
- ECE: average gap between the mean score and the observed rate over the 10 groups, weighted by group size.

Figure 1 shows that the row scores are clearly miscalibrated. Their mean is 0.434, while only 11.1% of the test stays are readmitted. Plus, the distribution histogram is centered around 0.4. The reliability curve lies below the diagonal over the whole range (for example, stays with a score around 0.7 are readmitted only 29% of the time). The Brier score (=0.2051) is even worse than the base-rate reference (0.0987). The ECE is 0.323.

Therefore, we can say that the model is overconfident, which means that it overestimates the risk at every level. This is a side effect of 'class_weight' being balanced. Readmitted stays count about 7.7 times more in the training loss; all scores are pushed up. The ranking of patients (PR AUC = 0.235) and the decision threshold, which was chosen for these scores, are not affected. However, we wouldn't trust the raw scores as probabilities for an important medical decision (for example, estimating the expected number of readmissions). This motivates the recalibration step described in the next part.

RECALIBRATION – PLATT scaling makes probabilities more reliable

To recalibrate the raw scores, we compared Platt scaling and isotonic regression. Both were fitted on the out-of-fold scores of the train set, and the choice was made on the train set only. Using the same 5 folds as before, each method was fitted on four folds and evaluated on the fifth. Our rule was to keep the simpler recalibration method unless isotonic regression was clearly better.

The two methods were almost identical (Brier = 0.09632 for Platt and = 0.09639 for isotonic, isotonic being better in only 1 fold out of 5). So we kept Platt scaling. The test set was then used to check this choice (Figure 2: Reliability curves and score distribution of the raw, isotonic, and Platt scores) :

- The mean probability (11.6%) now matches the real readmission rate (11.1%), and the reliability curve follows the diagonal
- The Brier score drops from 0.2051 to 0.0933 (better than the base-rate reference of 0.0987), and the ECE drops from 0.323 to 0.008
- The PR-AUC stays at 0.235, since the ranking is unchanged. Isotonic regression, by contrast, keeps only 78 distinct scores and lowers the PR-AUC to 0.228.

The calibrated probabilities can now be trusted, with two cautions: they rarely exceed 30%, because the model separates patients only moderately, and calibration must be monitored over time, since it can break if the patient population changes.

Calibrated probabilities also make the decision threshold interpretable. Indeed, our F1 threshold (0.542) corresponds to a 15% risk, which implicitly assumes that a missed readmission costs about 5.7 times more than an unnecessary follow-up. The threshold should be adjusted once the hospital defines its real costs.

CALIBRATION DIAGNOSTICS

FAIRNESS – Audit on "race": no evidence of unequal opportunity

We audited the model with respect to race. The model never uses this column, but other features can be correlated with it, so removing it does not guarantee fairness.

We compared three groups on the test set: Caucasian (75% of the stays), African American (18%), and "Other/unknown", which merges the small groups (only 126 readmitted stays). For each group, we computed the metrics behind the main fairness definitions:

- selection rate (share of patients flagged) for demographic parity;
- recall (share of readmitted patients flagged) for equal opportunity;
- FPR, together with recall, for equalised odds;
- precision for predictive parity.

When base rates differ between groups, these definitions cannot all be satisfied at once. We therefore chose one main criterion before looking at the results: equal opportunity. In our case, the costly error is to miss a patient who will be readmitted, and being flagged is a benefit. So every patient who will be readmitted should have the same chance of being flagged, whatever their race.

With one global threshold, the two main groups receive almost identical decisions (Figure 3: Selection rate, recall, FPR and precision per group with one global threshold (before mitigation)) :

- Recall is 44.9% for Caucasian and 43.9% for African American patients. This 1-point gap is smaller than its uncertainty (standard error of about 2.5 points);
- selection rate (24% vs 23.9%), FPR (21.4% vs 21.3%), and precision (20.8% vs 21.1%) are also very close.
- The disparate impact ratio is 0.996 between the two main groups and 0.830 across the three groups, both above the 0.8 alarm level.

The larger gaps across the three groups (for example, 6.1 points of recall) come from the small "Other/unknown" group, whose recall has a standard error of about 4.5 points. They are mostly noise. Overall, we found no evidence of unequal opportunity between Caucasian and African American patients.

MITIGATION – Per-group thresholds not retained

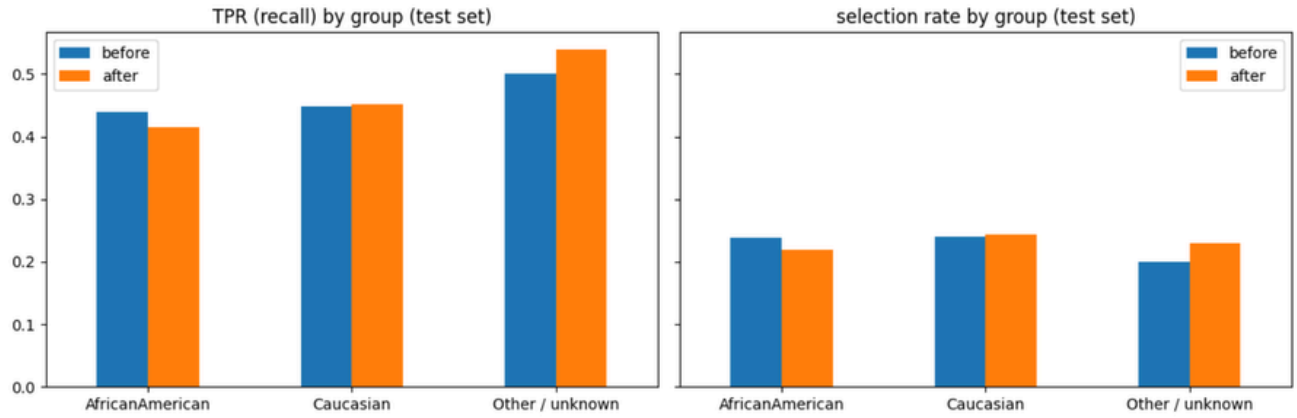
As a mitigation, we used one threshold per group and not one global threshold. Each threshold was chosen on the training set so that all groups reach the same recall. Then we compared the results on the test set (Figure 4).

We see that overall performance barely changed (F1 from 0.287 to 0.286), but the mitigation did not work:

- the recall gap across the three groups doubled (from 6.1 to 12.5 points);
- between the two main groups, the recall gap grew from 1.0 to 3.6 points, and the disparate impact ratio dropped from 0.996 to 0.901.

The initial gaps were already within noise, so the per-group thresholds mostly learned random variation from the training set. We therefore keep one global threshold. It is simpler and does not need the patient's race at decision time. We recommend monitoring recall by group over time instead.

Figure 4 : Recall and selection rate per group, before (=one global threshold) and after (=one threshold per group) mitigation



THE END

CONCLUSION

Early hospital readmissions are costly and are often a sign that the patient did not receive enough care. Using the Diabetes 130-US Hospitals dataset, we predict at discharge whether a diabetic patient will be readmitted within 30 days. Only about 11% of the stays are readmitted, so this was an imbalanced binary classification problem. A hospital could use such a model to offer extra follow-up to the patients most at risk.

We followed the full supervised-learning process:

- data exploration and a train/test split grouped by patient, to avoid leakage,
- feature engineering and class-imbalance handling,
- comparison of three models (Logistic Regression, Random Forest, LightGBM) in a single Pipeline, tuned with Optuna and tracked with MLflow,
- a final evaluation on an untouched test set.

We then explained the final model with SHAP, checked whether its probabilities can be trusted through calibration, and audited its fairness with respect to race.

CONTRIBUTION - Who did what?

	Main Task	**Writing**
Philomène	Production of the code and analysis	• Calibration • Fairness audit and mitigation • Conclusion, limits • Contribution statement and AI statement
Kelsy	Reviewed the code and add of comments	• Introduction • EDA, problem framing • Feature engineering • Explainability • Overall check of coherence and format of the document
Inès	Production of the code and analysis	• Metrics • Imbalance • The 3 models and the Pipeline • Optuna and MLflow • Final selection

AI - The usage

For this code, we first created a Google Document where we developed the framework for our work, taking content from the course slides and our notes. There, we outlined in detail every element we planned to include in the different sections of the project.

Here is an example of how we proceeded overall:
We created a detailed outline of what we wanted to analyze in the EDA, then asked an LLM to generate the specific code we needed, as we lack some advanced coding skills. When we didn't understand what the LLM was generating, we asked it to simplify it until we understood it completly, or to explain it to us like we were some five-year old kids. Once the EDA was built, we proceeded in the same way for the next section: determining which elements of the course to include in feature engineering based on what the EDA had taught us.
Therefore, we iteratively built the entire technical codebase with the help of an LLM, while remaining in control of the overall framework and the decisions to be made.

We wrote the analyses ourselves but some parts were translated using machine translation for better understanding and sentence flow.

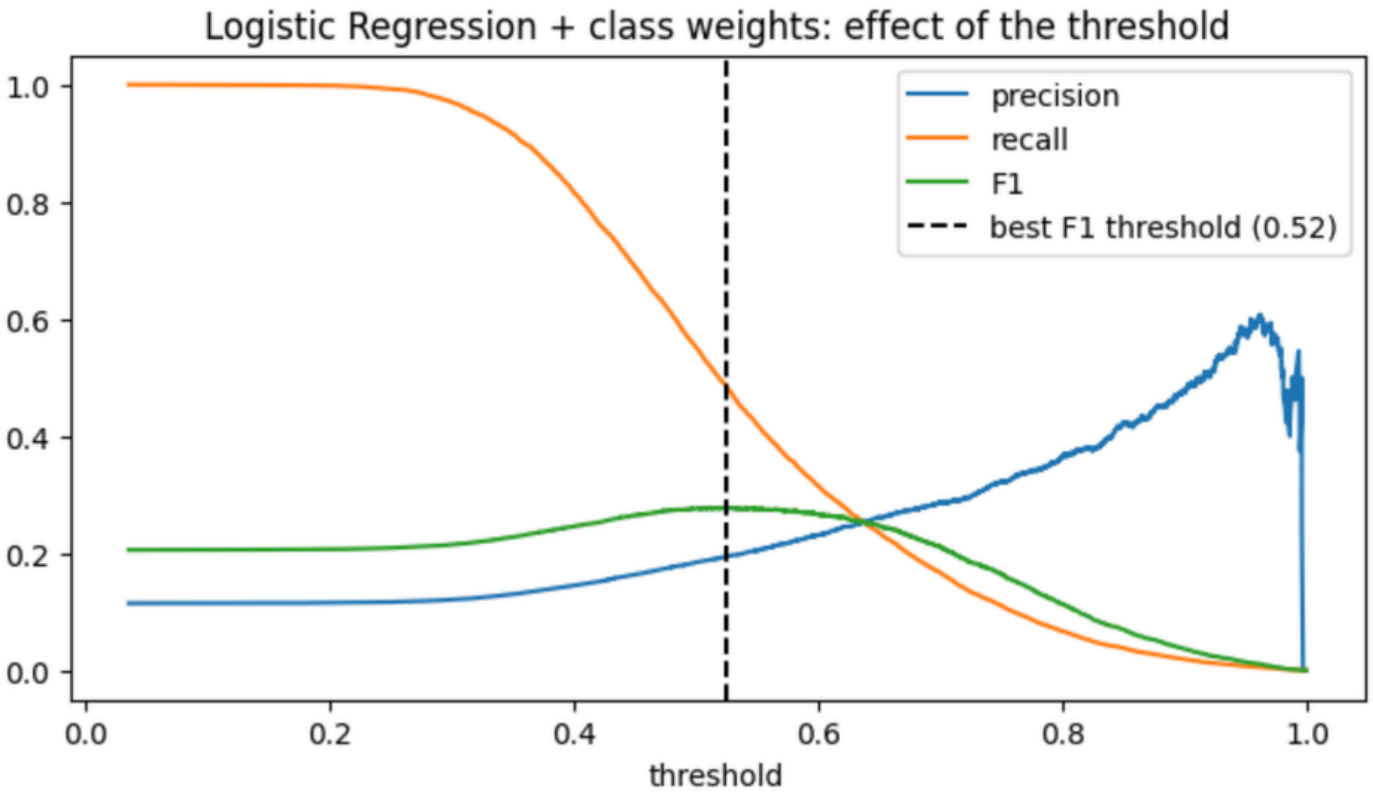
APPENDIX

CLASS IMBALANCE

cf. class_weights_test_results

	accuracy	precision	recall	f1	roc_auc	pr_auc
Dummy (always 0)	0.885	0.000	0.000	0.000	0.492	0.113
Logistic Regression	0.885	0.481	0.017	0.033	0.663	0.216
Logistic Regression + class weights	0.668	0.184	0.552	0.276	0.664	0.215

cf. class_weights_graph



cf. ablation_results

model	variant	PR-AUC	fold std	gain vs raw	std of the gain	folds where it helps
Logistic Regression	raw columns only	0.2142	0.0085	0.0000	0.0000	0/5
	+ n_prior_visits	0.2142	0.0086	-0.0000	0.0001	2/5
	+ n_med_changes	0.2142	0.0086	0.0000	0.0001	3/5
	+ age_years (replaces the age bracket)	0.2139	0.0086	-0.0003	0.0005	1/5
	+ diagnosis groups (replace ICD-9 codes)	0.2160	0.0095	0.0018	0.0019	4/5
	all of them (our pipeline)	0.2155	0.0095	0.0013	0.0021	3/5
LightGBM	raw columns only	0.2269	0.0116	0.0000	0.0000	0/5
	+ n_prior_visits	0.2280	0.0122	0.0011	0.0014	4/5
	+ n_med_changes	0.2278	0.0130	0.0009	0.0022	4/5
	+ age_years (replaces the age bracket)	0.2271	0.0129	0.0002	0.0019	2/5
	+ diagnosis groups (replace ICD-9 codes)	0.2290	0.0128	0.0021	0.0027	4/5
	all of them (our pipeline)	0.2301	0.0147	0.0032	0.0038	4/5

APPENDIX

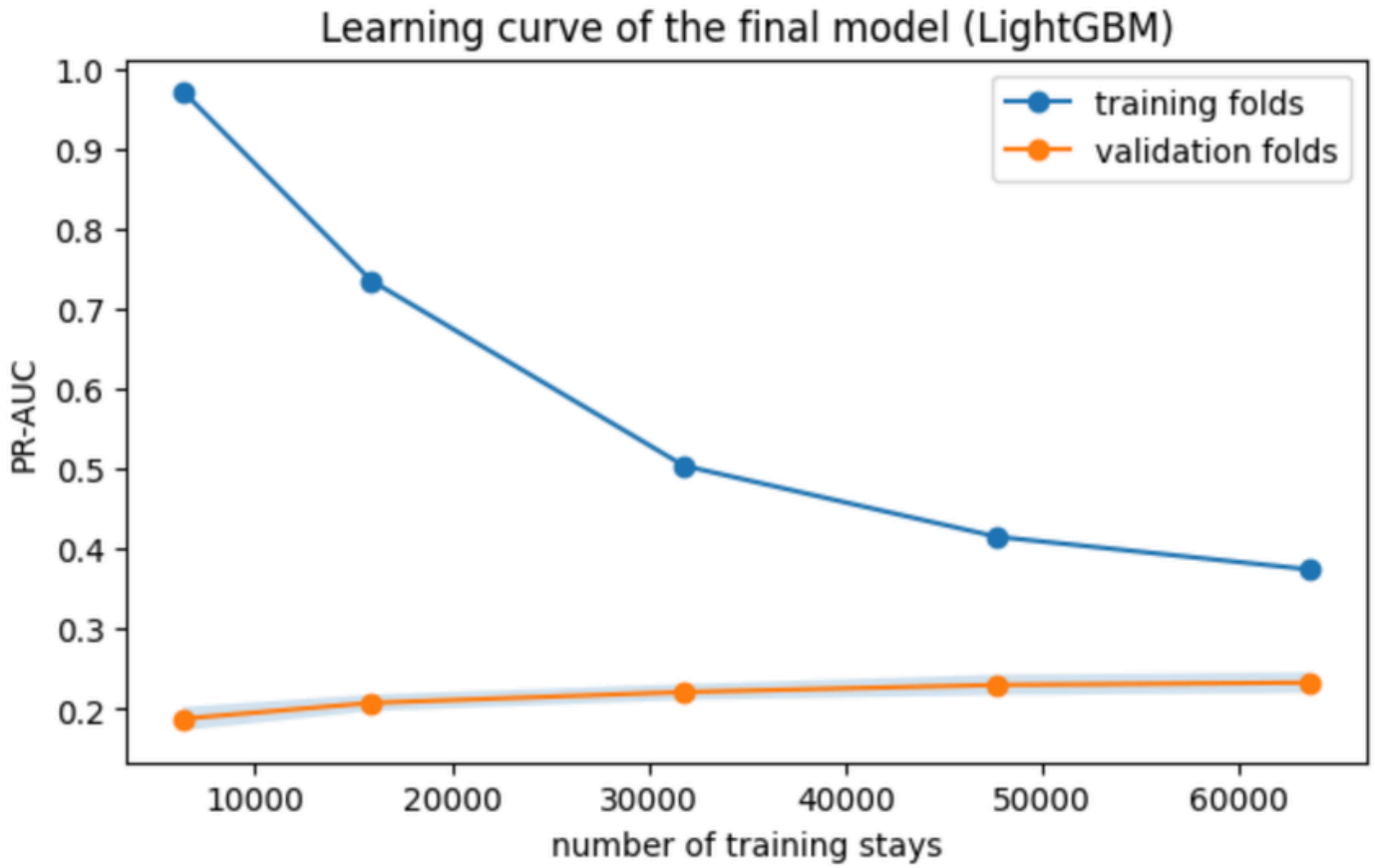
MODEL TUNING

cf. tuned_models_comparison_results

	accuracy	precision	recall	f1	roc_auc	pr_auc	best-F1 threshold	CV PR-AUC (mean of folds)
Logistic Regression	0.709	0.195	0.489	0.278	0.664	0.215	0.524	0.216
Random Forest	0.695	0.194	0.526	0.283	0.670	0.219	0.371	0.220
LightGBM	0.751	0.216	0.446	0.291	0.676	0.233	0.542	0.234

Final model: LightGBM
Threshold chosen on the train set: 0.542

cf. learning_curve_lightgbm



TESTING

cf. test_set_results

	test value	95% CI low	95% CI high
pr_auc	0.235	0.213	0.258
roc_auc	0.682	0.668	0.695
recall	0.450	0.425	0.476
precision	0.211	0.196	0.225
f1	0.287	0.269	0.305

APPENDIX

CALIBRATION

Figure 1 : Reliability curve and distribution of predicted scores before calibration.

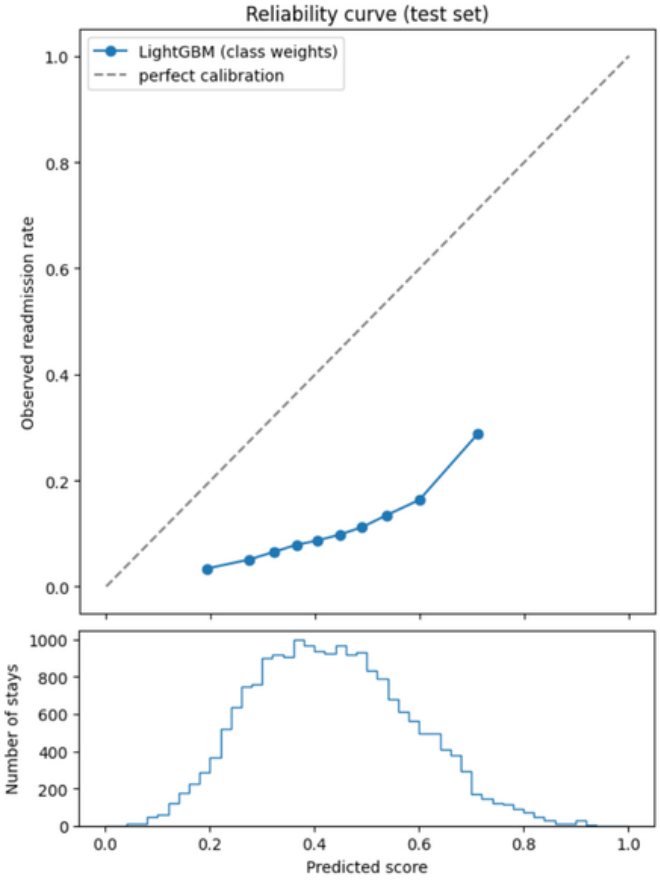
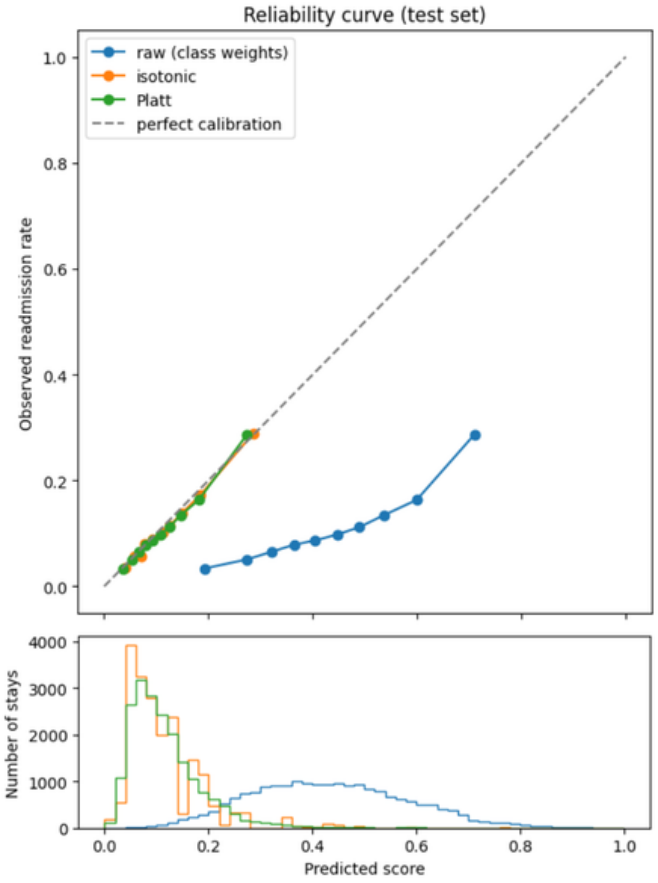


Figure 2 : Reliability curves and score distribution of the raw, isotonic and Platt scores.



FAIRNESS & BIAS AUDIT

Figure 3 : Selection rate, recall, FPR and precision per group with one global threshold (before mitigation)

	n_stays	n_readmitted	readmission rate	selection rate	TPR (recall)	TPR std (approx.)	FPR	precision
AfricanAmerican	3578.0	410.0	0.115	0.239	0.439	0.025	0.213	0.211
Caucasian	14825.0	1652.0	0.111	0.240	0.449	0.012	0.214	0.208
Other / unknown	1315.0	126.0	0.096	0.199	0.500	0.045	0.167	0.240

REFERENCES – Bibliography

[1] Diabetes 130-US Hospitals for Years 1999–2008, UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/296 (OpenML 4541).

[2] Strack B. et al. (2014), Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records, BioMed Research International, https://doi.org/10.1155/2014/781670.